MEHMET KIRTIŞOĞLU

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https://mkirtisoglu.github.io

Research Interest

- Algebraic (combinatorial) topology: Homotopy, homology, cohomology, and category theory.

Education

Bilkent University — Ankara, Turkey
Ph.D. in Mathematics, Advised by Prof. Dr. Ergün Yalçın — 2024–2028 (expected)

Bilkent University — Ankara, Turkey
M.Sc. in Mathematics, Advised by Prof. Dr. Ergün Yalçın — 2022–2024

- Master's Thesis: "Thomason's Homotopy Colimit Theorem and Cohomology of Categories"

Bilkent University — Ankara, Turkey
B.Sc. in Mathematics — 2018–2022

- Senior Project: "Reconstruction of Networks of Chaotic Systems", Advised by Prof. Dr. Fatihcan Atay

Publications

- Thomason Cohomology and Quillen's Theorem A, (with E. Yalçın) - Applied Categorical Structures 34, 22 (2026).

Scholarships and Awards

2024-2028	**National Ph.D. Scholarship Program (2211-A) by The Scientific and Technological Research Council of Turkey (TUBITAK)**
2022-2028	**Graduate Assistantship by Bilkent University**
2017-2022	**Outstanding Success Scholarship by Turkish Educational Foundation**
2017-2022	**National Undergraduate Scholarship Program (2205) by The Scientific and Technological Research Council of Turkey (TUBITAK)**
2017-2022	**Comprehensive Scholarship by Bilkent University**
2017	**Ranked in first 4000 in Nationwide University Entrance Exam among 2 million high school students in Turkey**

Organizing

- **Bilkent Mathematics Graduate Seminar Series** — Fall 2024-2025
  Graduate Student seminars series. The meeting aims to foster collaboration and discussion among graduate mathematics students of Bilkent University and to showcase the topics being studied in our department for undergraduate students.
- **IV. Mathematics Meeting, Middle East Technical University** — Oct 2018
  Student seminars series. The meeting is held at different universities each semester and five to six hundred students attend.
- **V. Mathematics Meeting, Istanbul Bilgi University** — March 2019
- **VI. Mathematics Meeting, Hacettepe University** — Oct 2019

Mentorship

- DRP Türkiye, Mentees: Ahmet Yusuf Öztürk, Boğaziçi University, on Homology, Aaditya Vicram Saraf, NISER, on Category Theory — Summer 2025

Workshops

- European Talbot Higher algebra and chromatic homotopy theory, Denmark with Gijs Heuts and Ishan Levy — 30 June - 4 July 2025
- Nesin Math Village Summer and Winter Schools, Turkey — 2016–2025
- Cakilarasi Math Village Topology of Surfaces, Turkey — 14 - 18 Aug 2023
- DRP Türkiye, on Homotopy Theory (mentored by Assist. Prof Feride Ceren Köse) — Summer 2022

Contributed Talks

- **Bilkent Mathematics Graduate Seminar Series** — Feb 2025
  "An Introduction to Cohomology of Small Categories"
- **Bilkent Mathematics Topology Seminars** — October 2023
  "Thomason's Homotopy Colimit Theorem"
- **Bilkent Mathematics Topology Seminars** — December 2022
  "Simplicial Quantum Contextuality"
- **DRP Turkey Symposium at Bogazici University** — Aug 2022
  "Seifert Van Kampen Theorem and Its Applications"
- **Bahar Mathematics Meeting VI, Hacettepe University** — October 2019
  "An Algebraic Topological Proof of the Fundamental Theorem of Algebra"

Academic Experience

Bilkent University — Ankara, Turkey
Instructor — Sep 2026 –present

- Math 106 (Calculus II for Business Administration)

Bilkent University — Ankara, Turkey
Graduate Assistantship, TA — Sep 2022 –May 2026

- Math 101 (Calculus I), Math 102 (Calculus II), Math 220/223/225/227 (Linear Algebra), Math 310 (Topology)

Network Oriented Dynamics and Data Science Research Group (NODDS) — Istanbul, Turkey
Internship at Kadir Has University — Nov 2020 –May 2022

- Worked on Graph Theory, Stability Theory, Synchronization Theory